

Solving Problems Involving Rates

Jonathan R. Bacolod

A **rate** compares two quantities with different units.

Speed

Speed is a rate that compares distance traveled to the time taken.

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Finding Distance

To find **distance**, multiply speed by time.

$$\text{Distance} = \text{Speed} \times \text{Time}$$

Finding Time

To find **time**, divide distance by speed.

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

Examples

Example 1

A car travels 180 km in 3 hours. What is its speed?

$$\text{Speed} = \frac{180}{3} = 60 \text{ km/h}$$

Example 2

A train moves at 80 km/h for 5 hours. How far does it travel?

$$\text{Distance} = 80 \times 5 = 400 \text{ km}$$

Example 3

A runner covers 15 km at a speed of 5 km/h.
How long does the run take?

$$\text{Time} = \frac{15}{5} = 3 \text{ hours}$$

Practice Exercises

Practice 1

A cyclist rides 72 km in 4 hours. What is the cyclist's speed?

Practice 2

A bus travels at 60 km/h for 7 hours. Find the distance traveled.

Practice 3

A delivery van travels 210 km at a speed of 70 km/h. How much time is needed?

Practice 4

A boat travels 96 km in 8 hours. Find its speed.

Practice 5

A motorcycle travels at 55 km/h for 6 hours.
How far does it travel?

Practice 6

A hiker walks 24 km in 6 hours. What is the hiker's speed?

Practice 7

A truck travels 450 km at a speed of 75 km/h.
How long does the trip take?

Activities

Activity 1

A jeepney travels 240 km in 4 hours. What is its speed in km/h?

Activity 2

A bicyclist rides 90 km in 5 hours. What is the bicyclist's speed?

Activity 3

A train moves at 90 km/h for 4 hours. How far does it travel?

Activity 4

A bus travels at 65 km/h for 8 hours. Find the distance traveled.

Activity 5

A runner covers 18 km at a speed of 6 km/h.
How long does the run take?

Activity 6

A delivery truck travels 320 km at a speed of 80 km/h. How much time is needed?

Activity 7

A ferry travels 126 km in 7 hours. Find its speed.

Activity 8

A motorcycle travels at 48 km/h for 5 hours.
How far does it travel?

Activity 9

A hiker walks 35 km in 7 hours. What is the hiker's speed?

Activity 10

A cargo truck travels 540 km at a speed of 90 km/h. How long does the trip take?

Thank you!